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Enhancing Marathi Abstractive Text Summarization with IndicBART

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Abstract—**The automatic text summarization technology is used in different industries such as news, law, medicine, education, social media, and business intelligence for their respective information extraction needs. Text summarization converts extended textual materials into shortened condensed content which maintains crucial points. It automatically produces summaries which retain essential sentences as well as all important facts found in the original document. The Text summarization field primarily focuses on English content but demonstrates limited research in Marathi language. Because of rich morphology and intricate syntactic patterns, Marathi language poses special difficulties for Natural Language Processing (NLP). Using pretrained transformer-based models, this study investigates Abstractive Text Summarization for Marathi with an emphasis on IndicBART, a language model created especially for Indian languages. Using a Marathi dataset, the IndicBART model has been refined to provide excellent abstractive summaries by utilizing its transfer learning and contextual awareness. The proposed approach is evaluated using ROUGE-1, ROUGE-2, and ROUGE-L metrics to measure its effectiveness**

Keywords-Extractive, Abstractive text Summarization (ATS), Extractive Text Summarization (ETS), IndicBART, Transformer Models, Natural Language Processing (NLP), Indic Languages

I. INTRODUCTION

Automated text summarization emerged as a vital research subject of Natural Language Processing because digital content gets expanded exponentially. The process of summarization produces condensed information from extensive body of text which enhances the efficiency of information retrieval systems. Summarization includes three different approaches that consist of Extractive , Abstractive and Hybrid Text Summarization. The extractive summation selects important text sections from the original document to create its final output. Abstractive summarization produces summaries which have a human-like quality because they both rephrase content and alter its structure. Hybrid summarization methods merge extractive and abstractive approaches to harness the advantages of both methods in order to create readable and comprehensive summaries[1].

The early research on Marathi text summarization chose extractive summarization as its main method by selecting important direct sentences from source texts. The methods of TextRank and Latent Semantic Analysis (LSA) operated as traditional approaches when seeking and extracting essential sentences from Marathi documents. Although useful these extraction methods produced results that lacked humanlike summaries because they depended on sentence selection rather than restructuring the text.

Marathi, a widely spoken Indo-Aryan language with over 83 million speakers, remains underrepresented in NLP research. Unlike English, Marathi has a rich morphology, free word order, and complex grammatical structures, making summarization a challenging task. Traditional rule-based and statistical methods fail to capture the semantic depth required for high-quality summarization. Text summarization performance has improved through deep learning and transformer-based models in numerous languages but Marathi language lacks well-trained models tailored for its linguistic characteristics.

Recent advancements in pretrained transformer-based models have shown promising results in NLP applications. Models such as BERT, mBART, T5, and IndicBART leverage transfer learning to generate contextually rich text representations. Among these, IndicBART, a variant of the BART model, is specifically designed for Indic languages and has demonstrated effectiveness in various text generation tasks. However, its application in abstractive text summarization for Marathi remains largely unexplored.

The research assesses the capability of IndicBART for abstract text summary generation in Marathi by using domain-specific training data for fine-tuning. ROUGE-1 joined by ROUGE-2, ROUGE-L and BLEU Score provide the assessment score for performance evaluation. Proposed study demonstrates the benefits of utilizing IndicBART, such as enhanced informativeness, coherence, and fluency in Marathi summaries.

The significant contributions of this paper is as follows

1)Improving IndicBART for Marathi abstractive text summarization.
2)Assessment of summary quality using ROUGE and BLEU measures.

Participating in this work aims to advance natural language processing techniques for Indian language which should result in better technologies for Indian languages. This research made use of Marathi training along with testing and validation data obtained from L3Cube-MahaSum and XLSum dataset. The proposed system implements IndicBART as the model for text summary generation. The IndicBART system operates as a sequence-to-sequence pre-training model which enables text generation across 11 Indian languages. The ROUGE and BLUE measurement provides accuracy assessment of created summaries. The assessment of the generated summary includes evaluation based on ROUGE-1,2, L and BLUE Score.

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II. LITERATURE REVIEW

IndicBART, a pre-trained sequence to sequence model supporting multilingual. The system has been specifically developed to operate with Indic languages which represent the spoken languages of most people in Maharashtra state of India. In addition to English the model provides supports for 11 Indian languages that include Hindi, Marathi, Assamese, Bengali, Gujarati, Punjabi, Kannada ,Oriya, Malayalam, Tamil and Telugu.

Authors R. Dabre et al. in their publication have compared IndicBART and mBART50 with each other for summarization task where the IndicBART performs slightly better for Marathi, Punjabi, Tamil and Telugu [2]. Authors P. D. Deshmukh et al.in this paper highlighted the potential of the IndicBART model and the importance of curated datasets in advancing Marathi text summarization research. The result shows 0.2432 as F1 score for ROUGE-1, 0.1711 for ROUGE-2 and 0.2392 for ROUGE-L [3]. Author S. D. Kale et al. introduced a new method to produce Marathi text document summaries through the combination of Natural Language Processing techniques together with deep learning models. The assessment method uses precision, recall and Mean Absolute Error (MAE) metrics. The research tests summary quality through an assessment composed of precision, recall and accuracy. The achieved precision in this approach amounts to 93.86% while recall reaches 95.51% and accuracy reaches 92.75% [4].

Authors Madhuri J N et al. in their study has provided a statistical method that ranks sentences according to their importance to produce extractive summaries of individual documents [5]. S. Deshpande et al. introduces a complete framework for handling multilingual together with cross-lingual text summary activities between Marathi and English documents. For English Summarization authors has used Pegasus, T5, and BART whereas For Marathi Summarization, IndicBART, mT5, and mBART models are employed. The Rouge-1 for IndicBART is 4.61, mT5-small is 5.3 and mBART-large-50 is 5.92[6].

In this paper author A. Chaudhari et al. presents a methodology for generating extractive summaries of Marathi text documents by leveraging neural network architectures. The proposed system operates by first translating the user's input Marathi text or documents into English. The system takes the English input text through translation before sending it to the summarizer to generate a summary which gets translated back to Marathi for user presentation. The study utilizes a dataset comprising 1,000 Marathi news articles. The system achieved a mean average precision of 0.319, recall value of 0.342 and overall accuracy of 0.323 [7]. The author Agarwal Arjit et al. presented a methodology for generating abstractive summaries of Hindi text by fine-tuning the IndicBART model The study utilized the Hindi language dataset provided by the ILSUM shared task. The obtained results are ROUGE-1 F1-Score: 0.544, Precision: 0.489 and Recall: 0.652 [8].

D. K. Gaikwad et al. presented a methodology for generating questions from Marathi text to aid in text summarization. This approach leverages a rule-based stemmer tailored for the Marathi language. The study utilized a dataset comprising 28 Marathi sentences selected to test the system's efficacy in generating questions. Accuracy of 60% is achieved [9]. Authors Apurva D. Dhawale et al. presents a methodology focusing on the pre-processing phase of Marathi text to enhance summarization efficiency. The researchers focus on the necessity of adapting preprocessing methods to handle the specific linguistic characteristics of Marathi to improve summarization performance [10].

Authors Apurva D. Dhawale et al. presented a system that provides students with a preparation solution through shorter Marathi e-news summary generation for competitive exams The system uses extractive text summarization to identify essential sentences from the source text which form the basis for summary content. To find and extract important sentences, it uses the TextRank algorithm which is unsupervised graph-based ranking model [11]. Sonali B. Kulkarni et al presents a paper in which it retrieve important lines from Marathi texts utilizing TextRank algorithm by analysing a dataset of 1,135 Marathi e-news articles which were obtained from GitHub [12]. Aparna Sawant et al. have proposed a system which introduces innovative LLM technology to develop Marathi text summaries. The research addresses NLP obstacles for Marathi as a low-resource language to enhance comprehension and accessibility of regional language information. This system generates a ROUGE-1 evaluation score at 65.4 and a ROUGE-2 result at 47.8 together with a ROUGE-L outcome at 62.1 [13]. V. V. Sarvadnya et al. has proposed a methodology for creating extractive summaries of Marathi text documents using a graph-based approach is presented in this paper. They have merged TextRank with word/thematic similarity and TextRank with a weighted positional distribution of sentence scores. The study makes use of a dataset that includes 634 Marathi news articles which includes politics, education, and current affairs. Using TextRank+ Positional ROUGE-1 F score was 0.8004 whereas using TextRank+ Similarity it was 0.8684 [14]. Authors Virat V Giri et al. presented a methodologies for extractive summarization of Marathi documents by employing Singular Value Decomposition inspired by Latent Semantic Analysis and fuzzy logic algorithms. The authors Virat V Giri et al. have developed a document extractive summarization approach that uses SVD from LSA together with fuzzy logic algorithms to process Marathi content.

Results obtained for single document summarization ROUGE1:F1 score using SVD was 0.612 and using Fuzzy logic was 0.623. Results obtained for multi-document summarization using SVD was 0.682 whereas using Fuzzy Logic was 0.63[15].

Authors Verma Pradeepika et al. in this paper presented the necessity for developing robust NLP tools tailored to Indian languages to enhance the performance of text summarization systems[16]. Author Awasthi Ishitva et al in this study tackled challenges in text summarization to enhance future models and incorporated reinforcement learning techniques to boost summarization efficiency. One identified limitation is the lack of sufficient labelled data for supervised learning methods [17]. Author Adhikari Surabhi examines different summarization processes to determine their success level through precision, recall, and F-measure metrics evaluation. Various techniques, including Machine Learning (ML), Neural Networks (NNs), Reinforcement Learning, Sequence-to-Sequence modelling, and Fuzzy Logic, have been applied to text summarization[18]. Authors V. V. Giri et al. offers an extensive evaluation of automatic text summarization

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systems which target the four Indian regional languages Bengali, Tamil, Kannada, and Punjabi. The Marathi text summarizer uses detailed explanation of its preprocessing methods according to the research paper [19]. Authors M. Gaikwad et al. in this survey paper provides overview of different categories of text summarization depending on various criteria like Classification according to input size, nature of output summary, Summary Language, Summarization Algorithm and summary content [20]. In this authors P. Kolhe et al have purposed the system using NLP and tools like WordNet for summarization. The system provides summaries from texts written in English, Hindi and Marathi languages [21]. Author Yogeshwari V. Rathod in this paper proposed system which has used Extractive text summarization approach and text rank graph based model to generate summaries of Marathi news article[22]. Authors Dhiraj Amin et al. in this research project centres around creating a question-answering system based on deep learning technology for Marathi language questions and responses. The research examines several transformer models particularly MuRIL, MahaBERT and IndicBERT after training them with multiple reading comprehension tasks from various Marathi text corpora. The testing phase demonstrated MuRIL as the superior model because it produced Exact Match score results of 0.64 alongside F1 score results of 0.74[23]. Authors Neha Rane et al. in the given paper investigates how different word embeddings perform in abstractive text summarization of Hindi and Marathi texts using ELMo, BERT and XLNet. The authors used ROUGE metrics to evaluate embedding methods by executing deep learning experiments across multiple datasets [24].

III. DATASET USED

A. The MahaSUM dataset: It is a large-scale collection of 25k Marathi news articles, curated to support abstractive summarization in Indic languages each with a headline, summary, and full text.

Dataset link is as follows:

Link: https://github.com/l3cube- pune/MarathiNLP/ tree/main/ L3Cube-MahaSum

The data is partitioned into three distinct sections of 22,000 training samples and 1,667 validation samples and 1,667 test samples for proper model learning and evaluation purposes.

B. XL-Sum dataset: BBC Research developed the large-scale multilingual summarization dataset XL-Sum that presents information across 44 languages with Marathi among them. The system enables human-written abstract text summaries that stay connected to the original context. This study utilized the Marathi subset from XL-Sum dataset for benchmark performance evaluation against IndicBART.

The dataset link is as follows:
https://huggingface.co/datasets/GEM/xlsum

The BBC Marathi news article collection provides 5,855 records for the Marathi language. The records contain the following fields
ID – A unique identifier for each news article.
URL – The link to the original BBC Marathi news article.
Title – Headline of the article.
Article – Full text of the news article.
Summary-Manually generated abstractive summary.
The dataset divides its contents into three sets as Training Set: 4,684 records, Validation Set: 732 records and Test Set: 439 records.

IV. PROPOSED METHODOLOGY

Fig 1 indicates the proposed methodology used for developing an abstractive text summarization system for Marathi language using the Indic BART model. It consists of steps like data collection, preprocessing, model selection and fine-tuning, and evaluation.

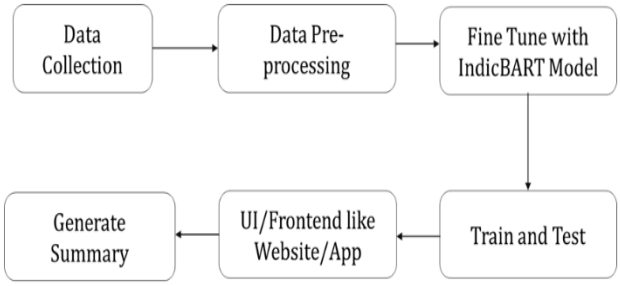

Figure 1: Proposed Methodology

A. Data Collection

The MahaSUM dataset features as a big-scale Marathi text corpus which focuses on developing Abstractive Text Summarization solutions for Indic language texts. A total of 25,000 Marathi news articles forms this resource which splits into headline, summary, and full text for training and testing purpose.

BBC Research developed the XL-Sum dataset as a large-scale multilingual summarization dataset which incorporates 44 languages with Marathi as one of them. The human-annotated summaries of high quality make XL-Sum a top-choice for research in Marathi text summarization.

B. Data Preprocessing

The method of converting raw data into a clean and structured arrangement before feeding it into a machine learning model. It involves the following techniques to improve model accuracy and efficiency.

1. Cleaning
2. Stop word Removal
3. Stemming & Lemmatization
4. Handling Encoding Issues
5. Text Normalization

1) Data Cleaning: It removes unwanted characters, punctuation, special symbols, and extra whitespace from the text. It helps in reducing noise in textual data.

2) Stop word Removal : Removing stop words stands as a fundamental stage in Natural Language Processing that truncates commonly used words which add no meaning value

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to analytical tasks including text summarization and sentiment analysis and machine translation. In Marathi stopwords include the common words "आहे", "आणि", "परंतु", "तर", and "यासाठी" that can reduce the meaningful content within sentences. The Indic NLP Library offers effective features for tokenization which supports processing of the Indian language Marathi. The Marathi stopwords are not included within the Indic NLP Library framework. The trivial tokenization process of the Indic NLP Library divides Marathi text into words and subwords while maintaining linguistic structure intact. The text is filtered by a predefined list of Marathi stopwords after performing tokenization. The system executes a sequential process on tokenized text while removing all words that correspond to the defined stopword list. The method keeps meaningful content while discarding useless information which enhances both processing speed and precision in subsequent NLP operations particularly text summarization tasks.

3) Stemming & Lemmatization: Both techniques reduce words to their root form to avoid redundant variations.
Stemming refers to the process where word reduction removes suffixes to create their base forms.

Example: "खेळला", "खेळत", "खेळेल" → "खेळ"
Lemmatization converts words to their root dictionary form while preserving meaning.

Example: "मुलींना" → "मुलगी", "मुलांची" → "मुलगा"

Lemmatization is more accurate than stemming but requires linguistic knowledge. The implementation of Indic WordNet leads to improved Marathi text summarization through effective lemmatization methods that convert words to base forms and maintain their meaning.

4) Tokenization: The text splitting operation known as tokenization produces smaller textual units called tokens from words and subword sequences down to single characters. Tokenization has significant impact on text summarization, named entity recognition, and machine translation. Tokenization task demonstrates complex difficulty when applied to multilingual data applications due to morphological elements.

System uses SentencePiece tokenizer compatible with the IndicBART model to handle Marathi text effectively.

SentencePiece is a language-independent tokenization technique where the sentence is split into subwords (pieces) using the shared vocabulary. The tokenization method follows a subword-based approach that efficiently handles complex linguistic structures and rich morphology found in Marathi.

5) Text Normalization: Standardizes text by handling spelling variations, number conversions, and abbreviation.

Example: Convert numbers to words (e.g., १२ → बारा).

Standardize spelling variations (माझ्या → माझे).

C. Model Selection

Selected IndicBART as a base model for text summarization task. IndicBART is a pretrained transformer model tailored for Indian languages. The model architecture of IndicBART consists of a six-layer encoder-decoder system which contains 244 million parameters to learn from data. IndicBART received training on the IndicCorp dataset so it could become more relevant and comprehensive for its operation across English and its 11 Indic languages. The IndicBART model shows better performance for low-resource Indian languages specifically designed for the Marathi summary task. The model enables text generation through translation, summary creation and text paraphrasing capabilities.

Loaded the pre-trained IndicBART model and tokenizer using the Hugging Face Transformers library.

D. Training Process:

1. Hyper parameter Tuning: The ability of the model to produce clear and compact summaries is improved through modifications of its hyper parameters which consist of Learning Rate, Batch Size, number of epochs, Max Input Length and Max Output Length. The model employs AdamW optimizer for better regularization and convergence.

2. Data Loading: Utilized DataLoader with appropriate batch sizes for training and testing datasets, employing shuffling for training data to improve generalization.

Batch data preparation: Batches = Dataset Size/Batch Size

3. Fine-Tuning the Model: Refine IndicBART model for Marathi dataset by training it to map input texts to their corresponding summaries.

The IndicBART model was trained to produce accurate summaries from input texts using pre-training on a specific Marathi dataset.

Applied techniques like gradient clipping to prevent gradients exploding and improve training stability.

E. User Interface Design:

Utilize the Gradio Framework to design an intuitive and user-friendly interface for ChatWhiz, incorporating input fields for different tasks.

F. Evaluation:

Most extensively used method for evaluating text summarization is ROUGE score. It computes the overlap of the produced summary with the reference summary using different n-gram and sequence-based approaches. The performance of summarization is assessed using ROUGE-1, ROUGE-2, ROUGE-L and BLUE score.

1) ROUGE-1:
Definition: This method determines the extent of shared unigram content between the produced summary and the benchmark summary.

Purpose: It offers a simple way to evaluate the level of overlap between specific terms in the benchmark summary and the produced summary.

Equation (1) shows ROUGE-1 calculation formula.

$$\text{ROUGE-1} = \frac{\text{Number of common single words}}{\text{Total unigrams in the reference summary}} \quad (1)$$

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Interpretation**:** A higher score indicates better content coverage.

2) ROUGE-2:

Definition: Evaluates how much the produced and benchmark summaries coincide in terms of bigrams, which are two-word sequences.

Purpose: By examining word sequences, it captures important elements of the summary generating process and offers insights regarding the coherence and fluency of the summary.

Equation (2) shows ROUGE-2 calculation formula.

$$\text{ROUGE-2}=\frac{\text{Number of common bigrams}}{\text{Total bigrams in the reference summary}} \quad (2)$$

Interpretation**:** By considering word pairs ,it mainly captures contextual coherence.

3) ROUGE-L:

Definition: The produced summary and the benchmark summary's longest common subsequence (LCS) match is measured.

Purpose: It assesses the summary's capability to maintain the structure and sequence of content as presented in the reference summary.

Equation (3) shows ROUGE-L calculation formula.

$$\text{ROUGE-L}=\frac{\text{Length of LCS}}{\text{Length of reference summary}} \quad (3)$$

Interpretation: Higher ROUGE-L indicates that the produced summary follows the structure of the benchmark summary more closely.

The evaluation of produced summaries uses three different ROUGE metrics known as ROUGE-1, ROUGE-2, and ROUGE-L. Equation (4,5,6) shows formula of Precision, Recall, and F1 Score. F1 Score contributes to the balance between precision and recall, giving a more comprehensive evaluation of summarization accuracy.

$$\text{Recall}=\frac{\text{Number of common unigrams}}{\text{Total unigrams in the reference summary}} \quad (4)$$

$$\text{Precision}=\frac{\text{Number of common unigrams}}{\text{Total unigrams in the generated summary}} \quad (5)$$

$$\text{Recall}=2*\frac{(precision*Recall)}{(Precision+Recall)} \quad (6)$$

4) BLEU Score: The BLEU metric serves as a common method for evaluating automated text which Papineni et al. presented in 2002 primarily for machine translation evaluation [25]. The similarity assessment between candidate and reference texts happens through n-gram precision calculation which analyzes single words and word sequences simultaneously. BLEU was initially created for machine translation, yet its developers expanded the algorithm for natural language processing applications that include text summarization. The BLEU evaluation method checks text summarization output for its similarity to reference summaries through n-gram analysis.

Table I shows the BLEU Score and its interpretation.

Table I: BLEU score range and its interpretation

BLEU score (%)	Interpretation
0-10	Poor
10-30	Fair
30-50	Good
50-70	Very Good
70-100	Excellent

V. RESULTS

Table II and III shows comparative analysis of proposed methodology along with mT5 and mBART models on MahaSUM and XLSum dataset by calculating ROUGE and BLEU score. The obtained BLEU score is 75.50 for MahaSUM dataset whereas 74.50 for XLSum dataset indicates that generated summaries are close to human written summaries.

Table II: Performance Analysis of different models on MahaSUM dataset

	IndicBART (fine-tuned)	mT5	mBART
ROUGE-1	91.7	87.5	86.4
ROUGE-2	89.8	85.8	84.7
ROUGE-L	86.7	83.4	82.4
BLEU Score	75.5	73.2	70.4

Table III: Performance Analysis of different models on XLSum dataset

	IndicBART (fine-tuned)	mT5	mBART
ROUGE-1	90.5	86.8	85.4
ROUGE-2	87.6	85.7	83.5
ROUGE-L	85.4	82.3	81.4
BLEU Score	74.5	72.2	70.4

Fig 2 shows the ROUGE and BLEU Score achieved by using fine-tuned IndicBART model along with mT5 and mBART model on the MahaSUM dataset whereas fig 3 shows on the XLSum dataset.

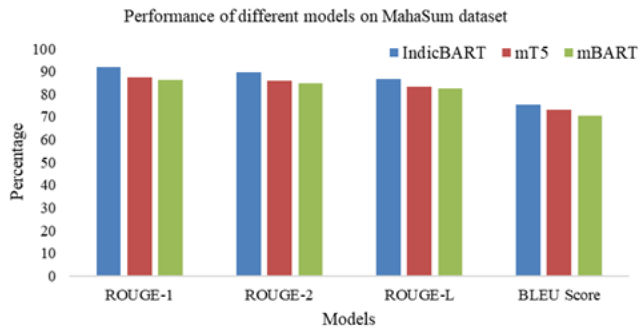

Figure 2:Comparative analysis of different models on MahaSUM dataset

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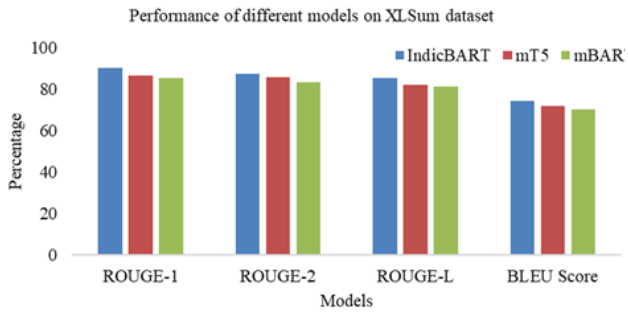

Figure 3: Comparative analysis of different models on XLSum dataset

The IndicBART system outperforms both mT5 and mBART during Marathi text summarization task. As trained on the high-quality MahaSUM dataset that includes extractive and abstractive summaries of Marathi news articles IndicBART generates superior scores according to both ROUGE and BLEU evaluations when compared against mT5 and mBART.

The IndicBART outperforms both mT5 and mBART when processing the BBC Marathi news articles in XL-Sum but showing relatively smaller improvement. The limited training record available in XL-Sum reduces its effectiveness compared to MahaSUM for deep-learning model training.

Fig 4 shows the GUI prepared for proposed methodology to generate summary for entered text.

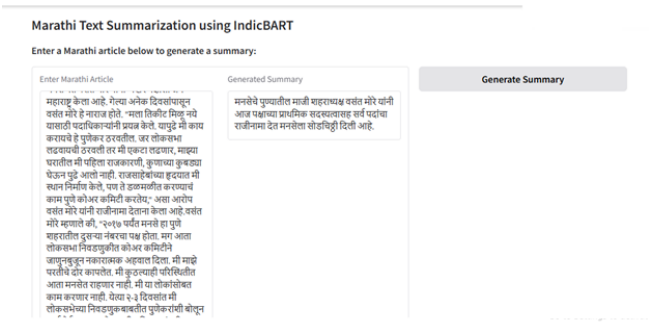

Figure 4: GUI for text Summarization

Table IV presents performance metrics of the fine-tuned IndicBART across four applications which include academic research abstracts, legal documents, social media posts and technical reports. The excellent fine-tuned IndicBART maintains superior performance for handling summarization task in various fields consistently.

Table IV: Use case dependent performance

Use Case	IndicBART (Fine Tuned) ROUGE-1
Educational Content	89%
Judicial Text Summarization	90%
Social Platform data	88%
Technical documents	87%

VI. CONCLUSION

This study indicates that IndicBART serves as an effective system for Marathi language summarization despite the scarcity of focus on low resource Natural Language Processing. The research examined IndicBART for Marathi abstractive text summarization and performed a performance evaluation against mT5 and mBART models. The outcome reveals that IndicBART achieves superior performance than mT5 and mBART especially after training with MahaSUM and XLSum dataset. The results show the ability of the IndicBART model to predict a high quality abstracts with good cohesion, as well as with respect to the context of the original text. Evaluation results using ROUGE and BLEU score affirms its potential in capturing the essence of input text while maintaining linguistic fluency.

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